Unit 8 Seminar: Machine Learning, Deep Learning and Ethical Impact

Introduction

Task 2 explains the model built in Task 3, so I take Task 3 first. Each section asks how far a reported figure can be trusted.

Comparing deep and conventional models (Task 1)

Yu et al. (2024) compare a CNN with four conventional models for predicting supply-chain costs. Their test table favours the CNN over the best alternative, a decision tree (RMSE 0.518 against 1.372), yet the abstract quotes the CNN's training figures (0.528, R^2 0.953) as test results. Computational efficiency is claimed without reported timings, and no prediction is explained. My team estimates, not forecasts, reviews per month, comparing a neural network with linear regression; forecasting would need dated outcomes and held-out future periods.

Building and testing a recommender (Task 3)

The supplied model gives each user and film a bias and hidden features. Its time-based split left 97 of 166 test users without training ratings (Table 1), so the test mostly measures cold start. Its advantage over a bias-only baseline was larger on users it knew (RMSE 1.023 against 1.119) than overall (1.040 against 1.045). Validation error rose after epoch 5 (Figure 1), the overfitting early stopping targets (Géron, 2025), yet the baseline won on validation (Table 2). Corrected to use all of a user's relevant films (Järvelin and Kekäläinen, 2002), the top-10 ranking score (NDCG at 10) fell from 0.304 to 0.149, below a most-rated list (Table 3).

Explaining its predictions (Task 2)

Here SHAP splits a surrogate model's near-copy of each score into contributions measured from the average prediction (Molnar, 2025). For Blade Runner, which user 13 had rated 4, the film's bias (+0.424) outweighed the user's strict ratings (−0.172; Table 4): correct arithmetic that says nothing about taste. A user with no ratings still gained +0.159, because a bias of 0 is above average. LIME or a chart would show the same unnamed terms (Molnar, 2025); named features (M and K.P., 2025) still need validating. As my team's lead and integrator, I would tell Airbnb's executives that the target is review activity, not bookings.

Conclusion

The figures were sounder than the claims built on them: most test users were new, a popularity list ranked better, and SHAP described arithmetic, not preference.

Evidence from my run

Tables 1 to 4 and Figure 1 come from my Colab run of 3 November 2026 at 14:47

UTC; Table 5 lists what I changed in the supplied notebook.

Who the test set contains

Model	All test ratings	Users seen in training	Users new to the model
Global mean	1.140	1.191	1.131
Bias baseline	1.045	1.119	1.032
Matrix factorisation, 30 epochs (supplied)	1.040	1.023	1.043

Table 1: Test RMSE by group of users, MovieLens 100K (Harper and Konstan, 2015). The supplied model predicts the average rating plus user and film biases plus the product of their hidden vectors. Of 166 test users, 97 had no training ratings; they gave 8,582 of the 10,000 test ratings. The model never saw these users, so their user bias stayed at 0 and their hidden vector at its random start. That may resemble a growing service, but it cannot show how well the model personalises.

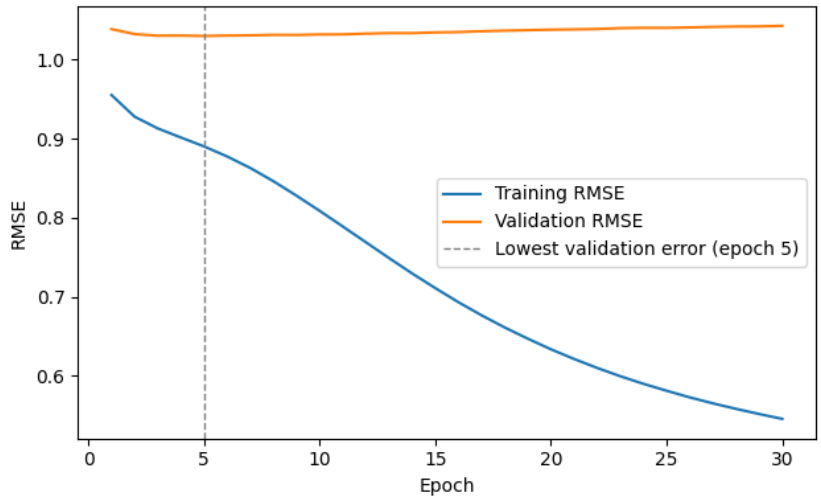


Figure 1: Training and validation RMSE over the 30 supplied epochs. Validation error is lowest at epoch 5 (1.0301) and 1.0429 by epoch 30, while training error falls from 0.955 to 0.546.

Accuracy and cost

Model	Validation RMSE	Test RMSE	Test RMSE, users seen	Time to fit
Bias baseline	1.019	1.045	1.119	0.014 s
Matrix factorisation, 30 epochs (supplied)	1.043	1.040	1.023	Not separately timed
Matrix factorisation, 5 epochs (early stop)	1.030	1.037	1.025	9.4 s

Table 2: Accuracy and fitting time. The 5-epoch model stops at the lowest validation error, chosen without the test set; it took 1.9 s per epoch. 0.25% of test predictions fell outside the 1 to 5 scale.

Ranking

Top-10 list	Users	Precision (top 10)	NDCG corrected	NDCG as supplied	Films covered
Matrix factorisation, all users	153	0.142	0.149	0.304	14.6%
Most-rated films, all users	153	0.190	0.209	0.411	4.5%
Matrix factorisation, users seen	57	0.058	0.079	0.161	14.3%
Most-rated films, users seen	57	0.091	0.104	0.207	4.5%
Matrix factorisation, new users	96	0.192	0.191	0.389	5.6%
Most-rated films, new users	96	0.249	0.271	0.532	0.6%

Table 3: Ranking (NDCG at 10) for the 153 evaluated users with a test rating of 4 or more, which counts as relevant. The most-rated list offers the most-rated films the user has not rated. The supplied cold-user slice (1 to 10 training ratings) held 1 user with 36 ratings. Results come from one seed and one split, without a significance test. New users keep random hidden vectors here, so the most-rated list beat this implementation, not collaborative filtering in general.

Explaining single predictions

Case	Prediction (surrogate)	SHAP: user bias	SHAP: film bias	SHAP: hidden terms	What the data show
Supplied example: user 90, Kundun (1997)	3.928 (3.929)	+0.159	+0.241	About +0.01	User 90 has no training ratings, so their bias is 0
Good film: user 13, Blade Runner (1982)	3.803 (3.804)	−0.172	+0.424	+0.033	User 13 rated it 4; average 4.12 over 223 ratings
Predicted low: user 13, Bio-Dome (1996)	0.301 (0.307)	−0.172	−1.681	−1.359	Average 1.89 over 28 ratings; below the scale

Table 4: SHAP on a ridge regression surrogate (Géron, 2025) of the supplied 30-epoch model. Its fidelity RMSE of 0.0027 is in-sample, on the 20,000 fitting rows; held-out fidelity was not measured. SHAP values add up to the surrogate's prediction from a base of 3.518. For Blade Runner the 32 hidden terms net +0.033, though single terms reach 0.36. User 13 has 608 training ratings, average 3.11 against 3.52 overall. The low film is user 13's lowest prediction among films with 20 or more training ratings, my threshold. User 90's hidden-term total is derived from the printed values. The supplied output (SHAP 0.50.0) gave the same 3.928 prediction different contributions (+0.177 and +0.400); my run used SHAP 0.54.1.

Changes to the supplied notebook

Type	In the supplied notebook	Change
Correction	NDCG at 10 built its ideal list only from hits already in the top 10.	The ideal list holds every relevant film the user has, up to 10.
Checks added	Asks about overfitting and for SHAP on a good and a low film, without code.	Test-set users, early stopping, a most-rated list, the cold-user slice, two SHAP cases.
Left as supplied	Unpinned installs and a SHAP deprecation warning.	Unchanged; package versions are printed in the run.

Table 5: The changes, also listed in the notebook's opening cell.

References

Géron, A. (2025) *Hands-On Machine Learning with Scikit-Learn and PyTorch*. Santa Rosa: O'Reilly.

Harper, F.M. and Konstan, J.A. (2015) 'The MovieLens datasets: history and context', *ACM Transactions on Interactive Intelligent Systems*, 5(4), article 19. Available at: https://doi.org/10.1145/2827872

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Yu, X., Tang, L., Long, L. and Sina, M. (2024) 'Comparison of deep and conventional machine learning models for prediction of one supply chain management distribution cost', *Scientific Reports*, 14, article 24195. Available at: https://doi.org/10.1038/s41598-024-75114-9

Supplied materials: University of Essex Online Unit 8 seminar brief and the notebook Recommender_exercise.ipynb.

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